221-2327 Sec-E M.Rohaan

Q#01:-

Sol.

$$\hat{P} = \frac{\text{no. of success}}{\text{sample size}}$$

$$= 8/12 = 0.67$$

Q#02:-

Sol.

$$Var(\hat{p}) = \frac{P(1-P)}{n}$$

$$= \frac{0.67(1-0.67)}{12} = 0.018$$

Q#03:-

Sol.

$$Var(X_1 + X_2 + X_3) = Var(X_1) + Var(X_2) + Var(X_3)$$

$$Var(X_i) = 9$$

$$Var(X_1 + X_2 + X_3) = 9 + 9 + 9 = 27$$

$$Var(\hat{\mu}) = Var\left(\frac{X_1 + X_2 + X_3}{3}\right) = \frac{1}{9} \cdot Var(X_1 + X_2 + X_3)$$

$$V(\hat{\mu}) = \frac{1}{9} \cdot 27 = 3$$

$$\hat{\mu} = 3$$

Q#04:-

Sol.

$$E(\hat{\mu}) = E\left(\frac{2X_1 + X_2}{3}\right)$$

$$= \frac{1}{3}\left(2E(X_1) + E(X_2)\right)$$

$$= \frac{1}{3}\left(2(\mu) + \mu\right) = \frac{1}{3} \times 3\mu = \mu$$

$$E(\hat{\mu}) - \mu = \mu - \mu = 0$$

Q#05:-

Sol.

$$Var(aX_1 + bX_2) = a^2 Var(X_1) + b^2 Var(X_2)$$

$$\hat{\mu} = \frac{2X_1 + X_2}{3}$$

$$Var(\hat{\mu}) = \left(\frac{2}{3}\right)^2 \times V(X_1) + \left(\frac{1}{3}\right)^2 V(X_2)$$

$$= \left(\frac{2}{3}\right)^2 \cdot 4 + \frac{1}{9} \cdot 9 \approx 2.78$$

Q#06:-

Sol.

$$V(\mu_1) = V\left(\frac{X_1 + X_2}{2}\right)$$

$$V(aX_1 + bX_2) = a^2 V(X_1) + b^2 V(X_2)$$

$$= \left(\frac{1}{2}\right)^2 \cdot V(X_1) + \left(\frac{1}{2}\right)^2 V(X_2) = \frac{1}{4} \cdot 10 + \frac{1}{4} \cdot 15$$

$$\approx 6.25$$

$$V(\mu_2) = V\left(\frac{3X_1 + X_2}{4}\right)$$

$$= \left(\frac{3}{4}\right)^2 \cdot 10 + \left(\frac{1}{4}\right)^2 \cdot 15 = 6.56$$

μ_1 has smaller variance

Q#07:-

Sol:

$$\hat{\mu} = \frac{10 + 14 + 12 + 8}{4} = 11$$

Q#08:-

Sol:

$$V(\hat{\mu}) = \frac{16}{4} = 4$$

Q#09:-

Sol:

$$MSE(\hat{\theta}) = V(\hat{\theta}) + (Bias(\hat{\theta}))^2$$

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta = 8 - 3 = 5$$

$$V(\hat{\theta}) = V(X) = 6$$

$$MSE(\hat{\theta}) = 6 + 5^2 = 31$$

Q#10:-

Sol:

$$MSE = V(\hat{\theta}) + (Bias(\hat{\theta}))^2$$

$$Bias(\hat{\theta}) = 0, \quad V(\hat{\theta}) = 2.5$$

$$MSE = 2.5 - 0^2 = 2.5$$